

AutoGuardian Pitch – Presentation Script

1. Title Slide

Good afternoon and thank you for giving me this opportunity.

My name is Nkosiyo Mdladla, and today I'm presenting an innovation I've been developing called AutoGuardian – a smart vehicle anti-theft system with real-time mobile integration.

2. The Problem

Vehicle theft remains a major issue in South Africa, with thousands of vehicles stolen every year.

Most current systems on the market are reactive – they only inform the owner after the theft has occurred.

Additionally, they are often too expensive for the average car owner, leaving many without protection.

This leaves a gap for a smart, low-cost, proactive solution that detects and prevents theft before it happens.

3. The Solution – AutoGuardian

AutoGuardian aims to solve this by combining sensor-based detection with GSM/GPS modules and mobile app controls.

The system alerts the owner in real-time when suspicious movement or entry is detected, and allows them to disable the car remotely using their phone.

It's affordable, effective, and scalable – ideal for personal cars, taxis, and fleet vehicles.

4. How It Works

The system is built around the ESP32 microcontroller, which communicates with:

- Motion and tilt sensors that detect unauthorized activity.
- A GSM module (e.g., SIM800L) that sends SMS alerts.
- A GPS module that provides real-time vehicle tracking.

All of this is controlled and monitored using a simple mobile application.

5. Key Features

Here's a summary of the key features:

- Real-time SMS or app-based alerts.
- Remote engine disable or ignition lock.
- GPS-based tracking for vehicle location.
- Integrated mobile interface for ease of use.
- Built using low-cost, widely available components.

6. Demo Preview

At this point in the presentation, I will show a demo.

DEMO INSTRUCTIONS:

- First, show the Wokwi simulation – how sensors trigger an alert and how the microcontroller responds.
- Second, show a screenshot or screen recording of the mobile app interface – highlighting how a user receives alerts and sends commands.
- If available, demonstrate how the GSM module sends SMS to a mobile phone.

7. Hardware Components

These are the main hardware components needed for the prototype:

- ESP32 Microcontroller
- SIM800L GSM Module
- GPS Module
- Vibration, tilt, or PIR motion sensors
- Power module, resistors, casing and jumper wires

I will show a detailed list to ensure full transparency of what will be purchased.

8. Estimated Budget

Here's a breakdown of the estimated costs for one complete unit:

- ESP32: R150 – R200
- GPS Module: R150 – R250
- GSM Module: R200 – R300
- Sensors: R100 – R150
- Miscellaneous (wires, casing): R100

Total per unit: Approximately R800 – R1,000

This makes it affordable and scalable, with high potential for rollout to many users.

9. Why Sponsor AutoGuardian?

AutoGuardian addresses a real and serious problem, with a working proof of concept.

It has strong commercial potential and real-world impact.

With funding and support, it can be refined and deployed more broadly.

I am committed to pushing this innovation forward with technical skill and passion, and I believe your support could make a meaningful difference.

10. Thank You

Thank you once again for the opportunity. I welcome your feedback and would be happy to answer any questions.